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(See 6.6)

MILITARY SPECIFICATION

MONITORS, VOLTAGE AND FREQUENCY, 400 HERTZ (HZ) ELECTRIC POWER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for voltage and frequency monitors for Naval shipboard installation.

1.2 Classification. The monitors shall be of the following types as specified (see 6.2.1):

Type VM1 - For sensing under voltage and over voltage limits on a single phase power system and providing direct current (dc) power to trip the connected remote feeder circuit breaker.

Type VM3 - For sensing under voltage and over voltage limits on a 3 phase power system and providing dc power to trip the connected remote feeder circuit breaker.

Type VFM3 - For sensing under voltage and over voltage and under frequency and over frequency limits on a 3 phase power system and providing dc power to trip the connected remote feeder circuit breaker.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests; H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915	- Cable and Cord Electrical, for Shipboard Use General Specification for.
MIL-C-915/30	- Cable, Electrical, 1000 Volts, Type TSGU.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036	- Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-O-16485	- Ohmmeters, Insulation-Resistance-Indicating Portable.
MIL-I-46058	- Insulating Compound, Electrical (for Coating Printed Circuit Assemblies).

STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-740	- Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipbuilding Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
DOD-STD-1399, Section 300	- Interface Standard for Shipboard Systems Electric Power, Alternating Current (Metric).

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2.1), a sample monitor shall be subjected to first article inspection and tests (see 4.3 and 6.4).

3.2 General requirements. The monitors shall be in accordance with the requirements of MIL-E-917 and the requirements of this specification, except as otherwise specified in the contract or order.

3.2.1 Painting. Parts not fabricated of a corrosion-resisting material or having a corrosion-resistant treatment as specified in MIL-E-917, except for semi-conductors, shall be painted as specified in MIL-E-917 except that only one coat of gray enamel shall be applied. Touching-up will be permitted for marks or scratches due to factory handling.

3.3 Operation. The monitors, when connected as shown on figure 1, shall sense the voltage and frequency conditions of all phases of the power source output whether connected 3 phase delta or wye (four wire grounded). In a delta connection, the phase sensing shall be between each line. In a wye connection, the phase sensing shall be between each line and the grounded neutral. The monitors shall operate within the specified voltage and frequency ranges specified in table I to provide tripping power to the circuit breaker when any one voltage or frequency exceeds the specified limit. When the monitor is connected to the output terminals of a power source there shall be no interaction between the control circuits of the monitor and the power source. On polyphase circuits the sensors shall respond to the highest root mean square (rms) value of the phase voltage irrespective of the values of the remaining phases. The performance of the voltage sensors shall be within the specified tolerances specified in table I, including the effects of line harmonics up to 4 percent per harmonic or 6 percent total. The line harmonics include all phase relationships relative to the fundamental. Frequency sensors shall operate within specified limits (see table I) under all voltage variations specified in table I and including the

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presence of harmonics as outlined above for the voltage sensors. Under the specified under voltage and over voltage and frequency conditions, the monitor shall provide a dc supply to trip the remote circuit breaker connected in the alternating current (ac) power circuit. Energy required to trip the circuit breaker shall correspond to 300 microfarad (μF) of capacitance charged to 125 volts dc. After the monitor has tripped the remote circuit breaker, the monitor shall automatically reset to permit closing of the circuit breaker by hand under favorable voltage and frequency conditions (see 3.4.6). The monitor shall not be damaged because of an open circuit or a ground located in the sensing leads or in the dc supply to the circuit breaker.

3.3.1 Input power. The monitor shall meet the performance requirements of 3.3 when powered in accordance with DOD-STD-1399, section 300, types II and III.

3.4 Electrical rating.

3.4.1 Sensing voltages (nominal). Each monitor shall operate on an input phase voltage of 115 volts delta, 200/115 volts wye, or 440 volts delta. The voltage application shall be determined at time of installation.

3.4.2 Sensing frequency (nominal). Each monitor shall operate on a phase input frequency of 400 Hz.

3.4.3 Monitor power type settings. The selection shall be set in the field between types II and III power in accordance with DOD-STD-1399, section 300, by operating an internal double throw switch. Monitor trip setting values shall be as specified in table I.

TABLE I. Monitor trip setting values (under all conditions of system phase voltage, frequency or current unbalance).

For	Type II power settings	Type III power settings
Over voltage	125 percent of nominal (± 5 percent)	115 percent of nominal (± 5 percent)
Under voltage	75 percent of nominal (± 5 percent)	85 to 90 percent nominal
Over frequency	430 Hz (± 5 Hz)	420 Hz (± 5 Hz)
Under frequency	370 Hz (± 5 Hz)	380 Hz (± 5 Hz)

3.4.4 Monitor setting trip delay. The trip values specified in table I shall be held from 45 milliseconds (ms) minimum to 70 ms maximum to prevent nuisance tripping because of possible system transients.

3.4.5 Trip energy for circuit breaker. The monitor circuit shall provide capacitor energy to trip the remote circuit breaker (200 feet maximum) two times on any one charge at all voltage and frequency conditions. The circuit breaker 120 volt dc shunt trip coils may have a resistance from 20 to 400 ohms.

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3.4.6 Trip energy for breaker (external signal). The monitor shall include two terminals for connection of external leads for tripping of the circuit breaker by the monitor when a closed contact signal is received from an external device such as a protective relay (see figure 1).

3.5 Static components. Field adjustment of the sensing circuits to compensate for aging of components shall be provided.

3.6 Duty. The monitor shall sense the power circuit continuously. The monitor shall provide the power to trip the breaker for 40 consecutive operations at 1-minute intervals (see 4.7.7).

3.7 Enclosure. The enclosures shall be of dripproof construction in accordance with MIL-E-2036 (see 4.7.1). Similar dimensions for the height and weight shall be provided. An identical enclosure shall be provided for type VM3 and VFM3 monitors. It shall be possible to remove or install any internal component without removing the enclosure from its mounting. Mounting means shall protrude from the sides of the enclosure. The mounting holes shall be suitable to take 5/16 inch screws. A wiring compartment approximately 2 inches high shall be provided at the bottom of the enclosure and a cable knockout of 1-1/32 inch diameter shall be placed on the two sides of the enclosure. In the bottom of the enclosure a 5/16 inch screw shall be provided for a condensation drain or as a breather hole.

3.8 Internal components. Components shall be accessible only when the front cover of the enclosure is removed. An internal single throw switch shall be provided for defeating the monitor trip circuit. The switch shall be labeled "SCR TRIP CIRCUIT" and the positions marked "ON" and "OFF". A second switch shall be provided for selection of the type power needed. The switch shall be labeled "TYPE POWER" and positions marked "II" and "III". The printed circuit board shall be coated in accordance with MIL-I-46058, types AR, SR, UR or XY.

3.9 Ventilation. Ventilation shall be by natural draft.

3.10 Environmental.

3.10.1 Ambient temperature. The normal ambient temperature shall be 50 degrees Celsius (°C). The monitor shall meet the requirements specified herein when operating over a range of 23 to 65°C.

3.10.2 Humidity. The monitor shall meet the requirements specified herein when exposed to an ambient humidity as specified in 4.7.14.

3.11 Mounting. The monitor shall be suitable for bulkhead mounting.

3.12 Size and weight. The size and weight of the equipment shall be held to a minimum.

3.13 Insulation resistance. The insulation resistance of all circuits shall be not less than 10 megohms when measured with all circuits both hot and cold (see 4.7.3).

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3.14 Dielectric strength. Monitors shall be constructed to withstand, for a period of 1 minute, a dielectric test voltage between separate circuits and to ground as follows (see 4.7.12):

Ac rms input or output voltage	Test voltage rms
60 and under	600
61 to 90	900
91 to 600	1000 plus twice rated
Over 600	2000 plus 2-1/4 times rated

3.15 Vibration. Monitors shall withstand type I vibration tests specified in MIL-STD-167-1 without mechanical damage or malfunctioning (see 4.7.13).

3.16 Shock. Monitors shall withstand the shock tests specified in MIL-S-901 on the light weight machine without the use of shock mounts. With rated voltage applied, all control devices shall function properly during and after the shock test. Mounting fixture 4A of MIL-S-901 shall apply (see 4.7.14).

3.17 Noise. Monitors shall meet the noise requirements specified in MIL-STD-740 for type 3, grade C equipment.

3.18 Electromagnetic interference (EMI). Monitors shall conform to the emission and susceptibility requirements and limits of MIL-STD-461 for class A4 surface ship (Servo/Synchro) equipment with the following addition:

RS03 limits

Frequency	E-field level (volts/meter)
14 KHz to 14 GHz	200

3.19 Inclined operation. Monitors shall show no change in operation when inclined at an angle of 45 degrees from the vertical in any direction (see 4.7.8).

3.20 Warm-up. Monitors shall meet the performance requirements specified herein after being energized for a maximum of 1 minute (see 4.7.5).

3.21 Component life. Monitors shall meet the specified 168-hour component life test (see 4.7.11).

3.22 Diagrams and identification plates.

3.22.1 Connection diagram. Each monitor shall contain a diagram of electrical connections for external wiring with each terminal and lead designated to correspond to markings as shown on the monitor. In addition, a schematic circuit, a wiring diagram, a description of operation and a list of recommended repair parts shall be included. Diagrams and instruction sheets shall be subject to approval upon request of the Naval Sea Systems Command (NAVSEA) and shall be secured and protected in accordance with MIL-E-2036. This information shall be sealed between sheets of clear plastic material and placed in a holder mounted on the inside of the front cover of the enclosure.

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3.22.2 Identification plate. Each monitor shall be provided with an identification plate in accordance with MIL-P-15024 and MIL-P-15024/5 located inside and fastened to the bottom of the enclosure. The plate shall be brass or corrosion-resistant steel not less than 0.030 inch in thickness and marked by etching, engraving, stamping or photoprocess. The marking shall be not less than 0.003 inch deep. The identification plate for the complete equipment assembly shall indicate the following data:

"MONITOR:_____VOLTAGE_____FREQUENCY_____AC_____

TYPE:_____PHASE_____DUTY: CONT.

INPUT VOLTAGE: 115/440 FREQUENCY: 400 Hz

OUTPUT: 120 dc AMBIENT: 50°C

CAT. NO._____SERIAL NO._____YEAR_____

STK NO._____CONTRACT:_____

MANUAL: NAVSEA_____ACCEPTANCE STAMP

MANUFACTURER'S NAME"

3.22.3 Component identification. Components shall be identified by stamping or stenciling the data on the component or the data may be located immediately adjacent to the component.

3.22.4 Warning plate. A warning plate shall be located within the enclosure indicating the following:

"WARNING
DISCHARGE CAPACITOR
BEFORE INSPECTION"

3.23 Technical data. When specified (see 6.2.1), the contractor shall prepare engineering drawings and technical manuals in accordance with the data ordering documents included in the contract (see 6.2.2).

3.24 Workmanship. Metal surfaces shall have a smooth finish and all details of manufacture, including the preparation of parts and accessories, shall be in accordance with the best practice for high quality electrical equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, lacquering, riveting, clearance between connections, ruggedness, and suitability of enclosure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Comparison inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 First article inspection. First article inspection shall be performed on one monitor of each type. First article inspection shall consist of the examination and tests specified in table II.

TABLE II. First article, comparison and quality conformance inspection.

Inspection	First article	Comparison	Quality conformance	Requirement paragraph	Inspection paragraph
General examination	X	X	X	3.24	4.6
Effectiveness of enclosure	X	---	---	3.7	4.7.1
Creepage and clearance	X	X	---	3.2	4.7.2
Insulation resistance	X	X	---	3.13	4.7.3
Wiring	X	X	---	3.3	4.7.4
Warm-up	X	X	---	3.20	4.7.5
Sensing	X	X	X	3.3	4.7.6
Trip duty	X	X	---	3.6	4.7.7
Inclined operation	X	X	---	3.19	4.7.8
Noise	X	X	---	3.17	4.7.9
EMI	X	---	---	3.18	4.7.10
Component life test	X	---	---	3.21	4.7.11
Dielectric strength	X	X	X	3.14	4.7.12
Vibration	X	---	---	3.15	4.7.15
Shock	X	X	---	3.16	4.7.16
Temperature	X	---	---	3.10.1	4.7.11
Humidity	X	---	---	3.10.2	4.7.14
Voltage surge	X	X	---	3.3.1	4.7.13

4.4 Comparison inspection. Two years after first article examination and test and after each 2 years in which the monitor is offered for delivery thereafter, one monitor of each type shall be subjected to the comparison inspection specified in table II.

4.5 Quality conformance inspection. Each monitor shall be subjected to the quality conformance inspection specified in table II.

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4.6 Examination. Each monitor shall be examined for compliance with the workmanship requirements.

4.7 Tests. Unless otherwise specified (see 6.2.1), all tests requiring electrical loading of equipment shall be performed using a resistive load for uniformity and convenience.

4.7.1 Effectiveness of enclosure. The equipment enclosure shall be tested in accordance with MIL-E-2036 for dripproof requirement. Dripproof enclosures shall be accepted or rejected based on inspection to ascertain that the enclosure performs its specified function.

4.7.2 Creepage and clearance. Creepage and clearance distances shall be demonstrated by actual measurement in accordance with MIL-E-917.

4.7.3 Insulation resistance. Insulation resistance shall be as specified in 3.13. The measurement of insulation resistance shall be made with all circuits of equal voltage above ground connected together. Circuits or groups of circuits of different voltage above ground shall be tested separately. Insulation resistance shall be measured with an insulation-resistance-indicating meter (500 volts dc) conforming to type GC of MIL-O-16485. The time of test voltage application shall be not less than 60 seconds. Measurements shall be made at any convenient ambient temperature with all circuits both hot and cold. The relative humidity and ambient temperature shall be recorded.

4.7.4 Wiring. The external sensing leads to the monitor and the dc supply leads to the circuit breaker shall be subjected to open circuit and grounding to determine that no damage occurs to the monitor as specified in 3.3.

4.7.5 Warm-up. When the monitor is subjected to nominal voltage and frequency at both 23°C and 65°C ambient, the equipment shall meet the requirements of 3.20.

4.7.6 Sensing. The performance of the sensing circuits shall be verified by using 3 phase test circuits. The 115 volt test circuit shall apply to both delta and wye monitor connections. The following sensing tests may be done during and as a part of the component life test (see 4.7.11).

4.7.6.1 Over voltage, 3 phase, balanced. With a balanced 3 phase input, the voltage shall be increased from the specified nominal value to 125 percent over voltage for type I and II power settings. For type III power setting, the voltage shall be increased from the nominal to 115 percent over voltage. The monitor shall trip within plus or minus 5 percent of the voltage values specified herein.

4.7.6.2 Over voltage, 3 phase, unbalanced. The test for over voltage, 3 phase, unbalanced shall be as follows: (a) On one phase maintain a nominal voltage and on the other phases increase nominal voltage to an over voltage of 125 percent plus or minus 5 percent for types I and II power settings and 115 percent plus or minus 5 percent for type III power setting. (b) Repeat the above tests for the remaining phases for all power setting types. (c) Repeat the above tests by only increasing one phase voltage at a time until a fault condition is reached. The monitor shall trip upon each application of over voltage.

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4.7.6.3 Under voltage, 3 phase, balanced. With a balanced 3 phase input, the voltage shall be increased from zero volts to a value 5 volts above the maximum specified under voltage trip value. The voltage shall then be decreased to 75 percent of nominal voltage for types I and II power settings and 88 percent of nominal voltage for type III power settings. The monitor shall trip within plus or minus 5 percent of the above voltage values.

4.7.6.4 Under voltage, 3 phase, unbalanced. The test for under voltage, 3 phase, unbalanced shall be as follows:

(a) With a balanced 3 phase input at nominal voltage, decrease the voltage on two phases to the under voltage values specified in table I.
(b) Repeat this test for each of the remaining phases in both power type settings.
(c) Repeat the test by decreasing one phase voltage at a time. The monitor shall trip upon each application of under voltage.

4.7.6.5 Trip energy (no voltage). The input power shall be set at nominal voltage. One lead, two leads and all three leads shall be disconnected. The monitor shall trip circuit breaker 120 volt dc shunt coils having a resistance value from 20 to 400 ohms under each condition. Under each condition the breaker shall trip two times on each capacitor charge. The cable to the circuit breaker shall be in accordance with type TSGU-3 of MIL-C-915 and MIL-C-915/30 of an approximate length of 200 feet.

4.7.6.6 Trip energy (under voltage). Balanced 3 phase voltage input shall be increased from zero to 50 percent nominal voltage. The monitor shall trip the circuit breaker shunt trip coil within 5 seconds after reaching the 50 percent voltage settings.

4.7.6.7 Over frequency. The frequency of the input circuit shall be raised and lowered to determine compliance with the upper and lower monitor trip points (plus or minus 5 Hz) specified in table I for each power type setting. The input circuit during the testing shall be set just below the minimum over voltage trip value as specified in table I.

4.7.6.8 Under frequency. The frequency of the input circuit shall be raised and lowered to determine compliance with the upper and lower monitor trip points (plus or minus 5 Hz) specified in table I for each power type setting. The input circuit during the testing shall be set just above the maximum under voltage trip values as specified in table I.

4.7.6.9 Monitor setting trip delay. The monitor shall be tested for both types of power settings selected.

4.7.6.9.1 Trip delay, over voltage. With the monitor energized at nominal 3 phase voltage, a 45 to 70 millisecond over voltage pulse shall be induced of approximately 135 percent of nominal voltage. At least four pulses shall be inducted at a rate no faster than each 10 seconds. The monitor shall close its circuit to trip the circuit breaker during each pulse period.

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4.7.6.9.2 Trip delay, under voltage. With the monitor energized at nominal 3 phase voltage, a 45 to 70 millisecond under voltage pulse of 65 percent of nominal voltage shall be induced on each phase separately. At least four pulses shall be induced at a rate no faster than each 10 seconds. The monitor shall close its circuit to trip the circuit breaker during each pulse.

4.7.7 Trip duty. The monitor shall be energized with trip voltage and nominal frequency for 20 minutes with the trip circuit terminal shorted. The equivalent of 50 feet of 18 gauge wire may be used to short the terminals. There shall be no damage to the monitor after this test.

4.7.8 Inclined operation. The monitor shall be tested for inclined operation at rated voltage and frequency. Inclination of the monitor in the following positions shall not affect the performance requirements of 3.3 and 3.19:

(a) Inclined 45 degrees forward.
(b) Inclined 45 degrees backward.
(c) Inclined 45 degrees to the right side.
(d) Inclined 45 degrees to the left side.

4.7.9 Noise. Noise tests shall be conducted in accordance with MIL-STD-740 for type 3, grade C equipment (see 3.17). The monitor shall be energized simulating actual operation. Short test reports as defined in MIL-STD-740 shall be furnished.

4.7.10 Electromagnetic compatibility (EMC). The monitor shall be tested in accordance with MIL-STD-462 for the requirements specified in 3.18.

4.7.11 Component life. The monitor shall be engerized at approximately nominal voltage and frequency at a minimum temperature of 65°C for 168 hours. During each 24 hour period, the monitor shall be tested for at least one sensing test (see 4.7.6.7). In case of a power failure to oven or equipment, the test may be resumed from point of failure.

4.7.12 Dielectric strength.

4.7.12.1 General. The dielectric strength test shall be made after all of the tests specified in 4.7.1 through 4.7.11 have been completed. The dielectric test shall be made upon the completely assembled equipment and not upon individual parts, except in the case of repair parts which require dielectric tests.

4.7.12.2 Test voltage. The frequency of the testing voltage shall be not less than 60 Hz and shall approximate a true sine wave. The value of test voltage shall be as specified in 3.14 and shall be applied continuously for a period of 1 minute. Rectifiers, bridges and Zener references shall be shorted out before applying the voltage.

4.7.12.3 Measurement of test voltage. The measurement of the voltage used in dielectric tests shall be made by the voltmeter method whereby the meter derives its voltage from the high-voltage circuit either directly or by means of a voltmeter coil placed in the testing transformer, or through an auxiliary ratio transformer.

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4.7.12.4 Points of application. The test voltage shall be successively applied between each electric circuit and all other electrical circuits and grounded metal parts not electrically connected to it.

4.7.13 Voltage surge. The indicator shall be subjected to an impulse test of 450 V with spikes to 2500 V in accordance with figure 2 of DOD-STD-1399, section 300.

4.7.14 Humidity. The monitor shall be exposed to an ambient humidity within the humidity range specified in MIL-STD-810 and shall perform all functions following testing.

4.7.15 Vibration. The monitor shall be subjected to type I vibration tests in accordance with MIL-STD-167-1 when energized at approximately nominal voltage and frequency (see 3.15).

4.7.16 Shock. Shock tests shall be conducted in accordance with MIL-S-901 for grade A, class I equipment (see 3.16). After the shock tests, the equipment shall be spot checked for the specified operation (see 3.3).

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Monitors.

5.1.1 Preservation and packaging. Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the contractor's commercial practice when such meets these requirements.

5.1.2 Packing. Packing shall be accomplished in a manner which will ensure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commerical practice when such meets these requirements.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination.

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5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the contracting activity (see 6.2.1).

6. NOTES

6.1 Intended use. The monitors should be connected to the output terminals of rotating or static units supplying delta or wye connected power to electronic equipment. Upon sensing unsuitable power conditions, the monitor should provide dc power to trip the feeder circuit breaker thereby isolating the electronic loads.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Nomenclature (monitor).
(c) Type required (see 1.2).
(d) When first article inspection is required (see 3.1).
(e) When drawings and technical manuals are to be provided (see 3.23).
(f) When tests requiring electrical loading should use a resistive load (see 4.7).
(g) Preservation, packaging, packing and marking, if other than specified in 5.1 (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.23	Drawings, engineering and associated lists	DI-E-7031	Level 2 Drawing number - contractor Design designation - contractor
3.23	Manual, technical, standard	DI-M-2044	Type I of MIL-M-15071

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Goverment under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Provisioning. Provisioning Technical Documentation (PTD), spare parts and repair parts should be furnished as specified in the contract.

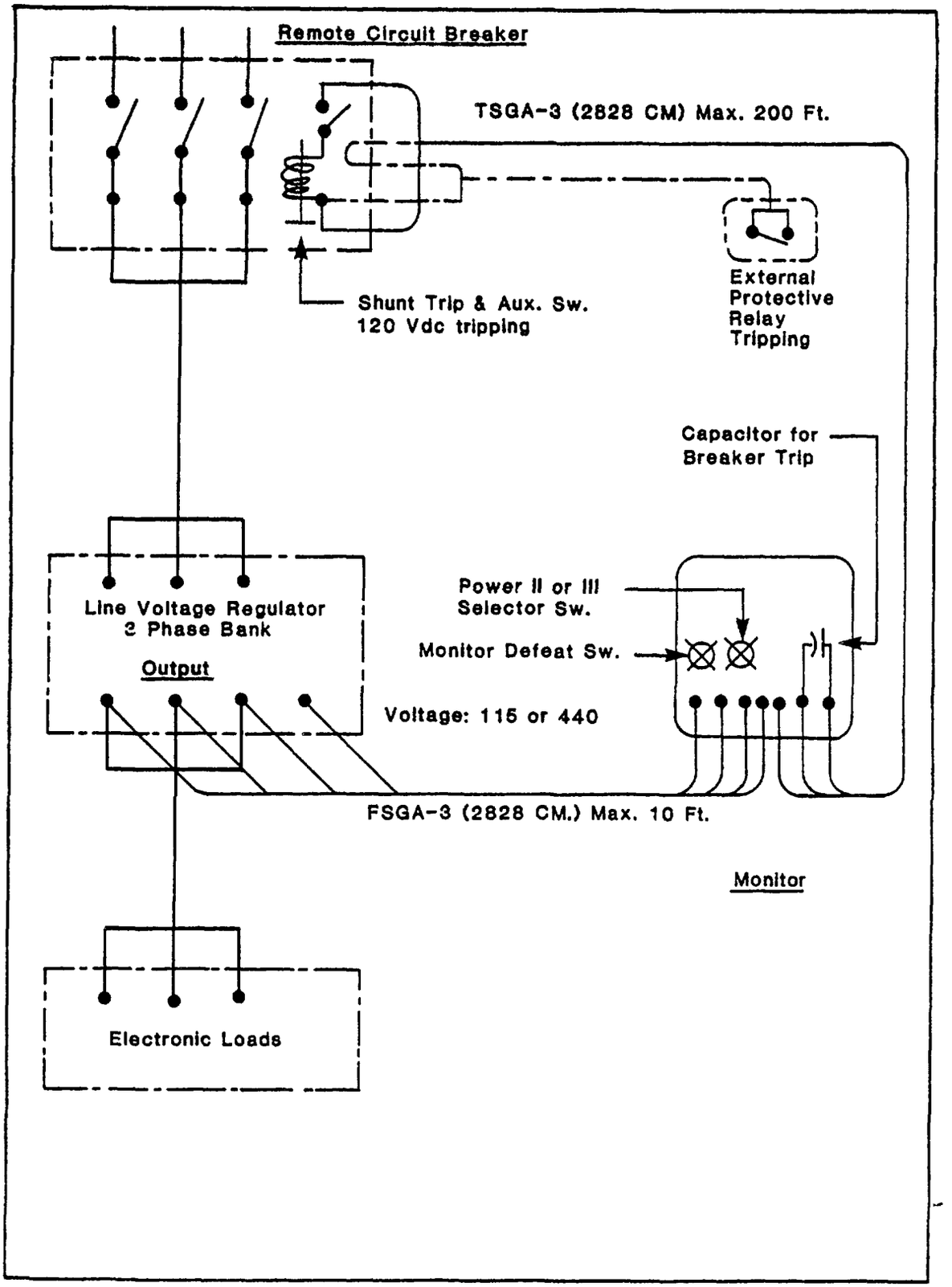
6.4 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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SH 131814

FIGURE 1. Typical external circuit for voltage and frequency monitors.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-24116B(SH	MONITORS, VOLTAGE AND FREQUENCY, 400 HERTZ (H7) ELECTRIC POWER

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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